



*AAM: Inefficient Campus
Resource Management
Solution: Student Eco Wallet*

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I. Abstract

Campus resource management plays an important role in maintaining a sustainable and environmentally responsible educational environment. However, many campuses experience problems such as leaking water taps, overflowing waste bins, malfunctioning lights, damaged infrastructure, and other issues that may remain unreported for long periods. These seemingly small problems can result in significant resource wastage and increased maintenance costs.

This report proposes a technology-based solution called the **Student Eco Wallet**, a digital platform that encourages students to actively participate in identifying and reporting resource related problems on campus. Through the platform, students can capture a photograph of an issue and upload it along with the location to the website. The student can receive Eco Points in their digital Eco Wallet as an incentive for contributing to sustainable campus management when the issue gets verified by the college staff. These points can be redeemed for approved campus benefits, such as discounts at the campus mess.

II. Introduction

Educational institutions consume a considerable number of resources every day to support students and faculty. Water, electricity, paper, food, and other materials are continuously used in classrooms, laboratories, offices, canteens, and libraries. Efficient management of these resources is therefore essential for reducing wastage, controlling operating costs, and creating a sustainable campus environment.

Despite the availability of maintenance departments and existing complaint mechanisms, resource-related problems on campuses are not always identified and addressed quickly. A dirty water filter that needs to be checked can take a long time. An overflowing bin may go unchecked. A faulty light, fan or broken equipment may not be fixed for a long period of time. Students and staff are often the first people to notice issues because they use campus facilities regularly. Information technology can be used to address this gap by creating a structured and accessible reporting system. A digital platform can allow students to report problems directly from their mobile devices, attach photographs and track the

progress of their reports. Such a system can also provide the administration of the various things that are happening around the campus without their knowledge.

Instead of treating students only as users who submit complaints, the system encourages them to become active participants in campus sustainability. By making reporting easier and rewarding meaningful contributions, the system can help create a green campus culture

III. Problem Statement

Campuses generate and consume significant amounts of resources every day, but the systems used to identify and address resource-related problems are often inefficient or dependent on manual reporting.

The absence of a centralized and location-based reporting mechanism can result in delays between the occurrence of a problem, its identification, and its resolution. This can increase resource wastage, reduce the efficiency of campus maintenance operations, and potentially increase operating costs. Normally complaint systems do not provide students with incentives to actively participate in identifying sustainability related issues.

Therefore, there is a need for an integrated digital system that enables students to easily report campus resource-related problems, provides accurate location information to the maintenance department responsible, allows the administration to track and manage reported issues and encourages students to help in the process.

The **Student Eco Wallet** is proposed as a solution to this problem. The system provides a structured platform through which students can report resource-related issues using photographs. Verified reports are converted into maintenance tickets and forwarded to the appropriate department, while students receive Eco Points for valid contributions. The proposed system aims to improve the speed and efficiency of issue reporting while encouraging students to actively contribute to sustainable resource management on campus.

IV. Sustainability Analysis

The Student Eco Wallet is a sustainability-oriented IT solution that aims to reduce campus resource wastage by enabling faster identification of problems and encouraging student participation. Its sustainability can be evaluated through the Sustainable Development Goals (SDGs), ISO 14001, Life Cycle Assessment (LCA), and Environmental Impact Assessment (EIA).

Sustainable Development Goals

The proposed solution is particularly relevant to **SDG 6, SDG 11, SDG 12, and SDG 13**, as it promotes water conservation, sustainable communities, responsible resource consumption, and environmental awareness.

SDG 6 – Clean Water and Sanitation

The system supports SDG 6 by allowing students to quickly report leaking taps, damaged pipes, and other water-related problems. Faster reporting can reduce the amount of water wasted before maintenance is carried out. The collected reports can also help the administration identify locations with recurring water-related problems and take preventive action.

SDG 11 – Sustainable Cities and Communities

A college campus functions as a small community with shared infrastructure and resources. The Student Eco Wallet encourages students to participate in maintaining their shared environment by reporting problems. The information collected can also help administrators identify recurring issues and areas that require improvement, contributing to a more sustainable campus.

SDG 12 – Responsible Consumption and Production

The system supports SDG 12 by helping identify unnecessary wastage of resources such as water, electricity, and materials. Reports of leaking taps, overflowing waste

bins, and unnecessary lighting can help the maintenance team take corrective action. The Eco Points system further encourages students to develop responsible resource-management habits.

SDG 13 – Climate Action

The Student Eco Wallet can indirectly contribute to climate action by improving resource efficiency. Reducing unnecessary water and electricity consumption can reduce the energy and resources required to supply them. The platform also increases student awareness of the environmental effects of everyday campus activities.

Relevance to ISO 14001

ISO 14001 provides a framework for organizations to establish and improve an Environmental Management System (EMS). The Student Eco Wallet can support such a system by providing a structured method for identifying environmental issues, recording them, assigning corrective actions, and monitoring their resolution.

For example, a report about a leaking tap can record its location, type, date, and status. Such information can later be analyzed to identify recurring problems and improve maintenance practices. However, the Student Eco Wallet itself would not make the institution ISO 14001 certified; it would function as a supporting tool within an environmental management system.

Life Cycle Assessment (LCA)

Since the Student Eco Wallet is primarily a digital service, its direct physical environmental impact is relatively low. However, its operation requires smartphones, network infrastructure, servers or cloud services, and electricity. These factors contribute to its environmental footprint.

At the same time, the system can generate environmental benefits by reducing resource wastage. For example, faster reporting of a leaking tap can reduce water loss, while improved waste reporting can prevent waste-management problems from becoming more serious. A detailed LCA could compare the environmental

impact of operating the platform with the resource savings achieved through improved campus management.

Environmental Impact Assessment (EIA)

A conventional EIA is generally used for projects such as large infrastructure or construction activities. Since the Student Eco Wallet is a software-based solution, a full-scale EIA would generally not be required. However, its environmental performance can still be monitored using indicators such as the number of issues reported and resolved, response time, recurring problems, and estimated resource savings.

Overall, the Student Eco Wallet supports resource conservation, responsible consumption, environmental management, and continuous improvement while encouraging students to actively participate in creating a more sustainable campus.

V. Proposed Engineering/IT Solution – Student Eco Wallet

The **Student Eco Wallet** is a digital platform designed to improve campus resource management by allowing students to report environmental and maintenance-related problems. It connects students with the appropriate maintenance department while rewarding valid contributions through an Eco Points system.

Working of the System

When a student notices an issue such as a leaking tap, overflowing waste bin, unnecessary lighting, or damaged facility, they can take a photograph and upload it through the platform. The system records the approximate location using the device's geolocation service. The report is then converted into a maintenance ticket containing the photograph, location, category, date, and time.

Based on the type and location of the issue, the ticket is automatically forwarded to the appropriate maintenance department. The department verifies the report, carries out the necessary action, and updates the ticket status. After a valid report is verified, Eco Points are credited to the student's digital wallet.

The basic workflow is:

Identify → Report → Geotag → Generate Ticket → Maintenance → Verify → Reward

Eco Points and Digital Wallet

The Eco Wallet acts as an incentive for student participation. Points are awarded only for genuine and verified reports to prevent false or duplicate submissions. The number of points can vary according to the usefulness of the reported issue.

Accumulated points can be redeemed for approved campus benefits, such as discounts at the campus mess. This provides students with a direct incentive while encouraging them to contribute to sustainable campus management.

Administrative System

A separate dashboard can be provided for maintenance staff and administrators. It can display reported issues and their status, such as New, Assigned, In Progress, Resolved, or Rejected. The dashboard can also provide information about frequently reported problems, response times, and locations with recurring issues.

The system can therefore help the campus move from reactive maintenance toward preventive maintenance by identifying recurring problems and supporting better resource-management decisions.

Main Technical Components

The proposed system can include a student web/mobile interface, geolocation services, a database for storing reports and Eco Points, a ticket-management system, notifications, and an administrative dashboard. Together, these components create a centralized platform for efficient campus resource management and student participation.

VI. Feasibility Analysis

The Student Eco Wallet is feasible as a campus-level IT solution because it can be developed using commonly available technologies and existing campus infrastructure.

Technical Feasibility

The system can be developed as a web or mobile application using standard technologies such as databases, cloud services, geolocation APIs, and notification systems. Most students already have smartphones with cameras and location services, reducing the need for additional hardware. The campus can also use an existing internet connection and computer systems for the administrative dashboard.

Economic Feasibility

The main costs would involve software development, hosting, database services, and maintenance. Since the system does not require extensive physical infrastructure, the initial investment can be relatively low. The reduction in resource wastage and improved maintenance efficiency could provide long-term financial benefits.

Operational Feasibility

The platform is designed to be simple for both students and maintenance staff. Students only need to photograph and submit an issue, while maintenance personnel can manage reports through a centralized dashboard. Training requirements would therefore be limited.

Challenges

Possible challenges include false or duplicate reports, misuse of Eco Points, privacy concerns related to location data, and low student participation. These can be addressed through user authentication, report verification, duplicate detection, appropriate privacy controls, and awareness programmes.

Overall, the Student Eco Wallet is technically, economically, and operationally feasible for implementation on a college campus.

VII. Expected Environmental Benefits

The Student Eco Wallet can provide several environmental benefits by improving the speed and efficiency of campus resource management.

- **Reduced water wastage:** Faster reporting of leaking taps, pipes, and other faults can reduce unnecessary water loss.
- **Improved waste management:** Reports of overflowing or improperly managed waste bins can enable quicker action and better waste segregation.
- **Reduced electricity wastage:** Reporting unnecessary lighting and electrical faults can help reduce avoidable energy consumption.
- **Increased environmental awareness:** The Eco Points system encourages students to actively participate in sustainable campus practices.

Overall, the system can help reduce resource wastage while promoting a culture of environmental responsibility among students and staff.

IX. Conclusion

Inefficient campus resource management can lead to unnecessary wastage of water, electricity, materials, and other resources. The Student Eco Wallet provides a technology-based approach to address this problem by allowing students to report resource-related issues using photographs and geolocation while automatically connecting the reports to the appropriate maintenance department.

The inclusion of Eco Points encourages students to actively participate in maintaining their campus environment, while the administrative dashboard can help identify recurring problems and support preventive maintenance. The solution is technically, economically, and operationally feasible and supports several sustainability objectives, including responsible resource consumption and environmental management.

Thus, the Student Eco Wallet can transform students from passive users of campus facilities into active participants in creating a more efficient and sustainable campus.

X. References

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